

ACADEMIC APPOINTMENTS

- **Nanjing University** Nanjing, China
Postdoctoral Researcher 2026–Present

EDUCATION

- **Nanjing University** Nanjing, China
Ph.D. - Meteorology Sep 2020 - June 2026
Advisor: Prof. Lili Lei
- **Nanjing University** Nanjing, China
B.S. - Atmospheric Science Sep 2016 - Jun 2020

RESEARCH INTERESTS

My research focuses on paleoclimate data assimilation (PDA), climate field reconstruction, and the integration of artificial intelligence (AI) with data assimilation. I am currently working on climate reconstructions of the Last Millennium to improve our understanding of climate variability and dynamics across multiple timescales.

1. Offline PDA: Paleoclimate data assimilation is most commonly implemented in an *offline* framework, where geological proxy records are combined with pre-computed climate model simulations. I developed an Integrated Hybrid Ensemble Kalman Filter (IHEnKF) that combines small-scale flow-dependent and large-scale climatological information, providing a computationally efficient framework while maintaining high reconstruction skill.

2. Online PDA: By replacing the numerical climate model with an AI-based surrogate model, I developed an *online* PDA framework based on IHEnKF that propagates proxy information forward in time. Furthermore, I proposed an Integrated Hybrid Ensemble Kalman Smoother (IHEnKS), which assimilates future proxies to further improve the reconstruction of past climate states.

PUBLICATIONS

- **Sun, H.**, L. Lei, Z. Tan, L. Ning, and Z. Liu, 2026 "An Ensemble Kalman Smoother for Online Paleoclimate Data Assimilation", *J. Adv. Model. Earth Syst.*, **18**, e2025MS005730, <https://doi.org/10.1029/2025MS005730>.
- Lei L., **Sun, H.**, Z. Tan, L. Ning, and Z. Liu, 2026 "Strongly Coupled Data Assimilation for Paleoclimate Reconstruction Using Deep Learning-Based Models.", *J. Adv. Model. Earth Syst.*, **18**, e2025MS005686, <https://doi.org/10.1029/2025MS005686>.
- **Sun, H.**, L. Lei, Z. Liu, L. Ning, and Z. Tan, 2025 "An Online Paleoclimate Data Assimilation with a Deep Learning-based Network", *J. Adv. Model. Earth Syst.*, **17**, e2024MS004675, <https://doi.org/10.1029/2024MS004675>.
- **Sun, H.**, L. Lei, Z. Liu, L. Ning, and Z. Tan, 2024 "A Hybrid Gain Analog Offline EnKF for Paleoclimate Data Assimilation", *J. Adv. Model. Earth Syst.*, **16**, e2022MS003414, <https://doi.org/10.1029/2022MS003414>.
- **Sun, H.**, L. Lei, Z. Liu, L. Ning, and Z. Tan, 2022 "An Analog Offline EnKF for Paleoclimate Data Assimilation", *J. Adv. Model. Earth Syst.*, **14**, e2021MS002674, <https://doi.org/10.1029/2021MS002674>.
- Z. Wang*, **H. Sun***, L. Lei, Z. Tan, and Y. Zhang, 2023 "The Importance of Data Assimilation Components for Initial Conditions and Subsequent Error Growth", *Sci China Earth Sci.*, **67**, 105-116, <https://doi.org/10.1007/s11430-023-1229-7>.
- K. Chen, Z. Liu, L. Ning, J. Liu, L. Lei, M. E. Mann, M. Yan, W. Sun, **H. Sun**, W. Hu, P. Gu, Y. Bao, Y. Qin, F. Xing, L. Li, J. Du, and Q. Wen, 2026 "Volcanic Eruptions Caused Weakening AMOC During the Preindustrial Past Millennium", *Nature Communications*, **17**, 8420, <https://doi.org/10.1038/s41467-026-74873-5>.
- F. Wu, L. Ning, Z. Liu, J. Liu, W. Hu, M. Yan, F. Xing, L. Lei, **H. Sun**, K. Chen, Y. Qin, B. Li, and C. Xu, 2025 "A New Last Two Millennium Reanalysis Based on Hybrid Gain Analog Offline EnKF and an Expanded Proxy Database", *npj Climate and Atmospheric Science*, **8**, 62, <https://doi.org/10.1038/s41612-025-00961-w>.
- W. Hu, L. Ning, Z. Liu, J. Liu, F. Wu, M. Yan, L. Jiang, L. Lei, F. Xing, **H. Sun**, K. Chen, Y. Qin, W. Sun, Q. Wen, and B. Li, 2024 "Reconstructing Tropical Monthly Sea Surface Temperature Variability by Assimilating Coral Proxy Datasets", *npj Climate and Atmospheric Science*, **7**, 261, <https://doi.org/10.1038/s41612-024-00816-w>.

* Denotes joint authorship.

PROJECT MANAGEMENT

- National Natural Science Foundation of China(Basic Research Program for Young Students), 2024-2026 "Ensemble-based Multi-scale Data Assimilation for Last Millennium Climate Reconstruction", **leader**
- National Key Research and Development Program of China, 2023-2028 "Data Assimilation Theory and Techniques for Holocene Climate Change in the Tropical Western Pacific", participation

CONFERENCES

- 8th WMO Symposium on Data Assimilation & 12th International Symposium on Data Assimilation (ISDA), Nanjing, China, Aug 31–Sep 4, 2026 (invited talk)
- European Geosciences Union (EGU) General Assembly, Vienna, Austria, May 3–8, 2026 (poster)
- 106th American Meteorological Society (AMS) Annual Meeting, Houston, USA, Jan 25–29, 2026 (poster)
- 11th International Symposium on Data Assimilation (ISDA), Melbourne, Australia, Sep 29–Oct 3, 2025 (poster)
- AGU23 Annual Meeting, American Geophysical Union (AGU), Online, Dec 11–15, 2023 (poster)
- 102nd American Meteorological Society (AMS) Annual Meeting, Online, Jan 23–27, 2022 (talk)

ACADEMIC SERVICE

- Reviewer, *Journal of Advances in Modeling Earth Systems (JAMES)*, 2026
- Reviewer, *Atmospheric Science Letters (ASL)*, 2025

TEACHING

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| • Teaching Assistant
<i>Graduate course: Data Assimilation</i> | Nanjing University
<i>Spring 2022</i> |
| • Teaching Assistant
<i>Undergraduate course: Dynamic Meteorology</i> | Nanjing University
<i>Fall 2021</i> |

AWARDS

- National Scholarship - 2024, CNY 30000
- Presidential Scholarship - 2020, CNY 20000
- Environmental Planning Scholarship - 2022, CNY 5000

PROGRAMMING

MATLAB/Python(numpy, scipy, xarray, Tensorflow, Pytorch)/Fortran/Linux/C